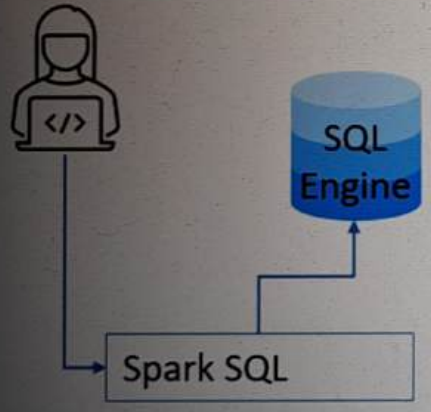
Understanding Spark Data Frame

Metadata Layer
Data Frame Schema

1. Where is the data for this table
2. What is the schema of the data and table

Meta Store OR Catalog



Text	Text	Text	Date	Int	Double	Text
InvoiceNo	StockCode	Description	InvoiceDate	Quantity	UnitPrice	Country
565231	47504K	ENGLISH ROSE GARDEN SECATEURS	01-06-2022 9.26	1	3.29	United Kingdom
565231	47566	PARTY BUNTING	01-06-2022 9.26	2	10.79	United Kingdom
565231	51014A	FEATHER PEN,HOT PINK	01-06-2022 9.26	2	0.83	United Kingdom
565231	72760B	VINTAGE CREAM 3 BASKET CAKE STAND	01-06-2022 9.26	1	20.79	United Kingdom
565231	72807A	SET/3 ROSE CANDLE IN JEWELLED BOX	01-06-2022 9.26	1	8.29	United Kingdom
565231	72807C	SET/3 VANILLA SCENTED CANDLE IN BOX	01-06-2022 9.26	1	8.29	United Kingdom
565231	82551	LAUNDRY 15C METAL SIGN	01-06-2022 9.26	1	2.46	United Kingdom
565231	82567	AIRLINE LOUNGE,METAL SIGN	01-06-2022 9.26	1	1.63	United Kingdom

Compute Layer
Spark Cluster

Physical Storage Layer
File in Distributed Storage

CSV, JSON, Parquet, AVRO, XML

Data File

ompute - Databricks Communi

community.cloud.databricks.com/?o=946780288864704#setting/clusters

Compute

All-purpose clusters | Job clusters

Create Cluster

All | Created by me | Accessible by me | Filter...

Name	State	Nodes	Runtime	Driver	Worker	Creator		Actions
demo-cluster	Terminated	-	9.1 LTS (includes Apache Spark 3.1.2, Scala 2.12)	Communit...	Communit...	prashant@sc...	0	...

1 - 1 of 1 | 20 / Page Go to 1

1. Cluster VM
2. Spark Metadata Store

Create Cluster - Databricks Com | 02-spark-dataframe-demo - Dat | 03-spark-table-demo - Databrick

community.cloud.databricks.com/?o=946780288864704#notebook/159151203742037/command/461968312746697

03-spark-table-demo (SQL)

demo-cluster

```
  OriginalPriority string,
  Priority string,
  FinalPriority integer,
  ALSUnit boolean,
  CallTypeGroup string,
  NumAlarms integer,
  UnitType string,
  UnitSequenceInCallDispatch integer,
  FirePreventionDistrict string,
  SupervisorDistrict string,
  Neighborhood string,
  Location string,
  RowID string,
  Delay float
) using parquet
```

Error running command 2. Go to command
Clear all notifications | Settings

1. Cluster VM (Compute Layer)
2. Spark Metadata (Metadata Layer)

Error in SQL statement: AnalysisException: Can not create the managed table(''demo_db`.`fire_service_calls_tbl`'). The associated location('dbfs:/user/hive/warehouse/demo_db.db/fire_service_calls_tbl') already exists.

Command took 2.16 seconds -- by prashant@scholarnest.com at 4/5/2022, 6:53:30 PM on demo-cluster

Cmd 3

```
insert into demo_db.fire_service_calls_tbl
values(1234, null, null, null, null, null, null, null, null, null, null, null, null, null, null, null, null, null, null,
null, null, null, null, null, null, null, null, null)
```

Command skipped

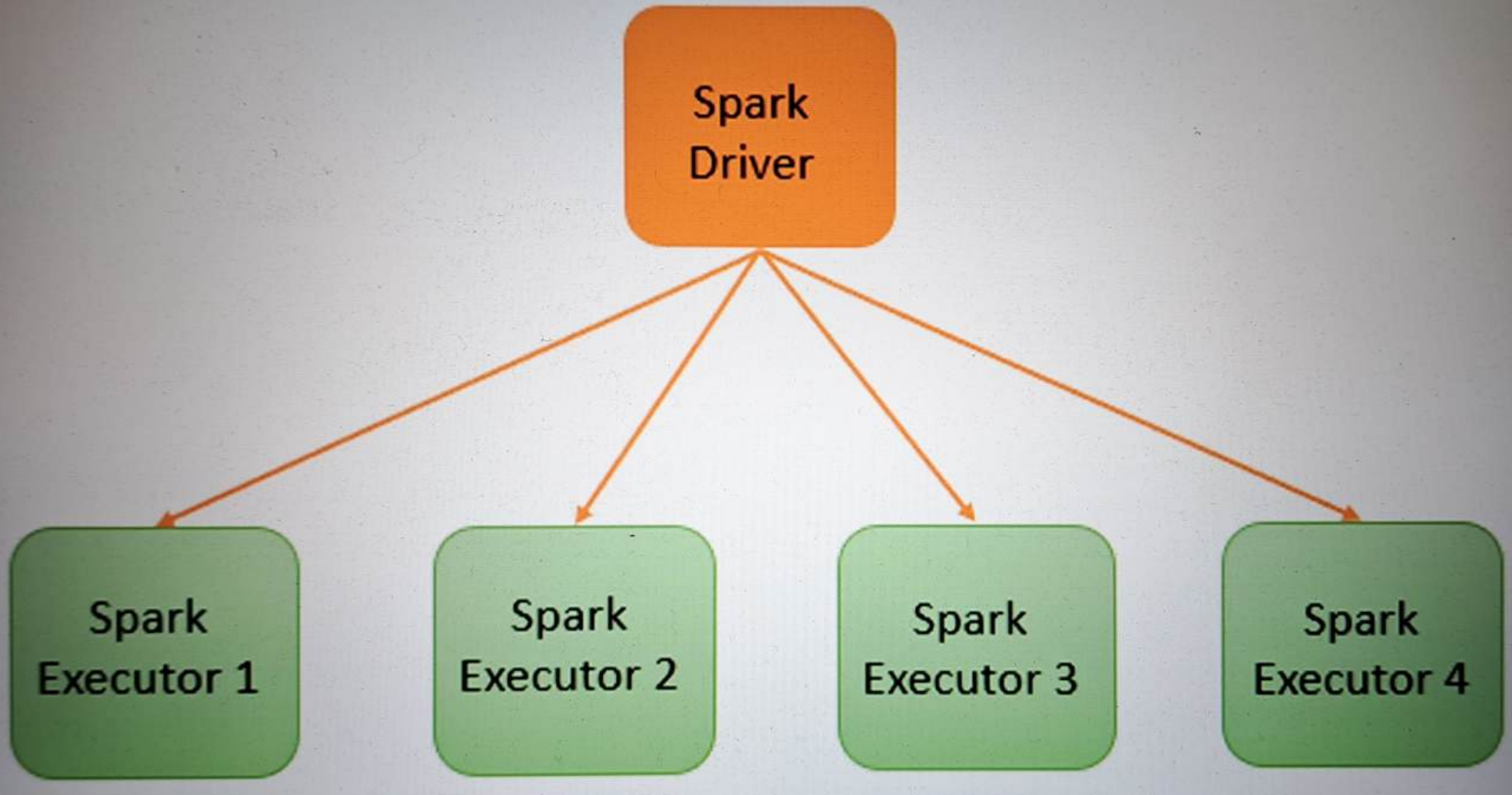
Command took 2.15 seconds -- by prashant@scholarnest.com at 4/5/2022, 6:53:30 PM on demo-cluster

Cmd 4

Dataframe Methods

1. Actions
2. Transformations
3. Functions/Methods

Spark Driver/Executor Architecture



Dataframe Methods

1. Actions

Actions are Dataframe operations that kick off a Spark Job execution and return to the Spark driver

2. Transformations

Spark Dataframe transformation produces a newly transformed Dataframe

3. Functions/Methods

Dataframe methods or functions which are not categorized into Actions or Transformations

Actions

1. collect
2. count
3. describe
4. first
5. foreach
6. foreachPartition
7. head
8. show
9. summary
10. tail
11. take
12. toLocalIterator

Actions

1. collect
2. count
3. describe
4. first
5. foreach
6. foreachPartition
7. head
8. show
9. summary
10. tail
11. take
12. toLocalIterator

Transformations

1. agg
2. alias
3. coalesce
4. colRegex
5. crossJoin
6. crosstab
7. cube
8. distinct
9. drop
10. drop_duplicates
11. dropDuplicates
12. dropna
13. exceptAll
14. filter
15. groupby
16. intintersect
17. ersectAll
18. join
19. limit
20. orderBy
21. randomSplit
22. repartition
23. repartitionByRange
24. rollup
25. sample
26. sampleBy
27. select
28. selectExpr
29. sort
30. sortWithinPartitions
31. subtract
32. transform
33. union
34. unionAll
35. unionByName
36. where
37. withColumn
38. withColumnRenamed

Udemy

Functions/Methods

1. approxQuantile
2. cache
3. checkpoint
4. createGlobalTempView
5. createOrReplaceGlobalTempView
6. createOrReplaceTempView
7. createTempView
8. explain
9. hint
10. inputFiles
11. isLocal
12. localCheckpoint
13. persist
14. printSchema
15. registerTempTable
16. toDF
17. toJSON
18. unpersist
19. writeTo
20. withWatermark

1. corr
2. cov
3. freqItems
4. mapInPandas
5. replace
6. sameSemantics
7. semanticHash
8. to_koalas
9. to_pandas_on_spark
10. toPandas

working-with-dataframe - Da × | 03-spark-table-demo - Databrick × | +

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/384359114294302

05-working-with-dataframe Python

demo-cluster

Command took 0.28 seconds -- by prashant@scholarnest.com at 4/7/2022, 1:37:17 PM on demo-cluster

Cmd 4

```
1 display(renamed_fire_df)
```

1. Read your raw data into a dataframe.
2. Transform your raw dataframe and create a new transformed Dataframe

(1) Spark Jobs

	CallNumber	UnitID	IncidentNumber	CallType	CallDate	WatchDate	CallFinalDisposition	AvailableDtTm	Addre
1	20110014	M29	2003234	Medical Incident	01/11/2002	01/10/2002	Other	01/11/2002 01:58:43 AM	10TH S
2	20110015	M08	2003233	Medical Incident	01/11/2002	01/10/2002	Other	01/11/2002 02:10:17 AM	300 Bl
3	20110016	B02	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:47:00 AM	2000 B
4	20110016	B04	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:51:54 AM	2000 B
5	20110016	D2	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:47:00 AM	2000 B
6	20110016	E03	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:47:00 AM	2000 B
7	20110016	E38	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:51:17 AM	2000 B

Truncated results, showing first 1000 rows.

Click to re-execute with maximum result limits.

Command took 0.76 seconds -- by prashant@scholarnest.com at 4/7/2022, 1:37:50 PM on demo-cluster

Cmd 5

Python

```
1
```

rking-with-dataframe - Da × 03-spark-table-demo - Databric × +

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/384359114294302

05-working-with-dataframe Python

demo-cluster

Command took 1.83 seconds -- by prashant@scholarnest.com at 4/7/2022, 11:32:44 AM on demo-cluster

Cmd 3

```python
renamed_fire_df = raw_fire_df \
    .withColumnRenamed("Call Number", "CallNumber") \
    .withColumnRenamed("Unit ID", "UnitID") \
    .withColumnRenamed("Incident Number", "IncidentNumber") \
    .withColumnRenamed("Call Date", "CallDate") \
    .withColumnRenamed("Watch Date", "WatchDate") \
    .withColumnRenamed("Call Final Disposition", "CallFinalDisposition") \
    .withColumnRenamed("Available DtTm", "AvailableDtTm") \
    .withColumnRenamed("Zipcode of Incident", "Zipcode") \
    .withColumnRenamed("Station Area", "StationArea") \
    .withColumnRenamed("Final Priority", "FinalPriority") \
    .withColumnRenamed("ALS Unit", "ALSUnit") \
    .withColumnRenamed("Call Type Group", "CallTypeGroup") \
    .withColumnRenamed("Unit sequence in call dispatch", "UnitSequenceInCallDispatch") \
    .withColumnRenamed("Fire Prevention District", "FirePreventionDistrict") \
    .withColumnRenamed("Supervisor District", "SupervisorDistrict")
```

1. You can create a chain of Spark transformation methods one after the other.
2. Spark transformation returns a new dataframe after transforming the old dataframe.
3. Spark dataframe is immutable.

Command took 0.28 seconds -- by prashant@scholarnest.com at 4/7/2022, 1:37:17 PM on demo-cluster

Cmd 4

```python
display(renamed_fire_df)
```

(1) Spark Jobs

5-working-with-dataframe - Da × 03-spark-table-demo - Databrick × +

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/384359114294301

05-working-with-dataframe Python

demo-cluster

Command took 0.28 seconds -- by prashant@scholarnest.com at 4/7/2022, 1:37:17 PM on demo-cluster

Cmd 4

```
1 display(raw_fire_df)
```

Spark DataFrame transformation produces a newly transformed dataframe.

(1) Spark Jobs

	Call Number	Unit ID	Incident Number	CallType	Call Date	Watch Date	Call Final Disposition	Available DtTm	Add
1	20110014	M29	2003234	Medical Incident	01/11/2002	01/10/2002	Other	01/11/2002 01:58:43 AM	10TH
2	20110015	M08	2003233	Medical Incident	01/11/2002	01/10/2002	Other	01/11/2002 02:10:17 AM	300
3	20110016	B02	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:47:00 AM	200C
4	20110016	B04	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:51:54 AM	200C
5	20110016	D2	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:47:00 AM	200C
6	20110016	E03	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:47:00 AM	200C
7	20110016	E38	2003235	Structure Fire	01/11/2002	01/10/2002	Other	01/11/2002 01:51:17 AM	200C

Truncated results, showing first 1000 rows.

Click to re-execute with maximum result limits.

Command took 0.56 seconds -- by prashant@scholarnest.com at 4/7/2022, 1:41:01 PM on demo-cluster

Cmd 5

Python

```
1 display(renamed_fire_df)
```

(1) Spark Jobs

-working-with-dataframe - De | 03-spark-table-demo - Databric | 04-spark-sql-demo - Databricks

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295

05-working-with-dataframe Python

demo-cluster

```python
fire_df.printSchema()
```

```
root
 |-- CallNumber: integer (nullable = true)
 |-- UnitID: string (nullable = true)
 |-- IncidentNumber: integer (nullable = true)
 |-- CallType: string (nullable = true)
 |-- CallDate: date (nullable = true)
 |-- WatchDate: date (nullable = true)
 |-- CallFinalDisposition: string (nullable = true)
 |-- AvailableDtTm: timestamp (nullable = true)
 |-- Address: string (nullable = true)
 |-- City: string (nullable = true)
 |-- Zipcode: integer (nullable = true)
 |-- Battalion: string (nullable = true)
 |-- StationArea: string (nullable = true)
 |-- Box: string (nullable = true)
 |-- OrigPriority: string (nullable = true)
 |-- Priority: string (nullable = true)
 |-- FinalPriority: integer (nullable = true)
 |-- ALSUnit: boolean (nullable = true)
 |-- CallTypeGroup: string (nullable = true)
 |-- NumAlarms: integer (nullable = true)
```

Command took 0.03 seconds -- by prashant@scholarnest.com at 4/7/2022, 3:18:42 PM on demo-cluster

Cmd 11

Column Transformation Methods:

1. withColumnRenamed
2. withColumn

Utility Method:

1. printSchema()

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/3271264990413480

05-working-with-dataframe Python

demo-cluster

Show result

Cmd 13

```
q1_df = fire_df.where("CallType is not null") \
            .select("CallType") \
            .distinct()
print(q1_df.count())
```

Actions are DataFrame operations that kick off a Spark Job execution and return to the Spark Driver.

(3) Spark Jobs

32

Command took 2.00 seconds -- by prashant@scholarnest.com at 4/8/2022, 12:34:24 AM on demo-cluster

Cmd 14

```
q1_df1 = fire_df.where("CallType is not null")
q1_df2 = q1_df1.select("CallType")
q1_df3 = q1_df2.distinct()
print(q1_df3.count())
```

(3) Spark Jobs

32

Command took 2.79 seconds -- by prashant@scholarnest.com at 4/8/2022, 12:37:20 AM on demo-cluster

Cmd 15

Markdown

Q2. What were distinct types of calls made to the Fire Department?

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/3271264990413480

05-working-with-dataframe Python

demo-cluster

Show result

Cmd 13

```
q1_df = fire_df.where("CallType is not null") \
            .select("CallType") \
            .distinct()
print(q1_df.count())
```

(3) Spark Jobs

32

Command took 2.00 seconds -- by prashant@scholarnest.com at 4/8/2022, 12:34:24 AM on demo-cluster

Cmd 14

```
q1_df1 = fire_df.where("CallType is not null")
q1_df2 = q1_df1.select("CallType")
q1_df3 = q1_df2.distinct()
print(q1_df3.count())
```

(3) Spark Jobs

32

Command took 2.79 seconds -- by prashant@scholarnest.com at 4/8/2022, 12:37:20 AM on demo-cluster

Cmd 15

Markdown

Q2. What were distinct types of calls made to the Fire Department?

Transformation Methods:

1. where()
2. select()
3. distinct()

Action Method:

1. count()

-working-with-dataframe - Da

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/3271264990413480

05-working-with-dataframe Python

demo-cluster

Show result

Cmd 13

```
q1_df = fire_df.where("CallType is not null") \
            .select("CallType") \
            .distinct()
print(q1_df.count())
```

(3) Spark Jobs

32

Command took 2.00 seconds -- by prashant@scholarnest.com at 4/8/2022,

Cmd 14

```
q1_df1 = fire_df.where("CallType is not null")
q1_df2 = q1_df1.select("CallType")
q1_df3 = q1_df2.distinct()
print(q1_df3.count())
```

(3) Spark Jobs

32

Command took 2.79 seconds -- by prashant@scholarnest.com at 4/8/2022, 12:37:20 AM on demo-cluster

Cmd 15

Markdown

Q2. What were distinct types of calls made to the Fire Department?

1. Analyzing the Dataframe:
   1. Convert your dataframe to a temporary view and Run your SQL on the view
   2. Use Spark Transformation and Action methods

2. Chain and Series of Transformation

3. Action Method - count()

-working-with-dataframe - Da

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/3271264990413481

05-working-with-dataframe Python

demo-cluster

Command took 1.93 seconds -- by prashant@scholarnest.com at 4/8/2022, 1:20:37 AM on demo-cluster

Cmd 15

Q2. What were distinct types of calls made to the Fire Department?

```
select distinct CallType as distinct_call_types
from fire_service_calls_tbl
where CallType is not null
```

Cmd 16

```
q2_df = fire_df.where("CallType is not null") \
            .select(expr("CallType as distinct_call_type")) \
            .distinct()
q2_df.show()
```

Actions are DataFrame operations that kick off a Spark Job, execute and return to the Spark Driver.

Cmd 17

Q3. Find out all response for delayed times greater than 5 mins?

```
select CallNumber, Delay
from fire_service_calls_tbl
where Delay > 5
```

1. Execute your code by triggering a Spark job.
2. Return the results to the driver
3. Show the results to you

Cmd 18

-working-with-dataframe - Da × pyspark.sql.GroupedData — PyS × +

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/3271264990413485

05-working-with-dataframe Python

demo-cluster

Cmd 20

Q4. What were the most common call types?

```
select CallType, count(*) as count
from fire_service_calls_tbl
where CallType is not null
group by CallType
order by count desc
```

Cmd 21

Python

```
fire_df.select("CallType") \
    .where("CallType is not null") \
    .groupBy("CallType") \
    .count() \
    .orderBy("count", ascending=False) \
    .show()
```

1. DataFrame.count() --> Action
2. GroupedData.count() --> Transformation

Cmd 22

Q5. What zip codes accounted for most common calls?

```
select CallType, ZipCode, count(*) as count
from fire_service_calls_tbl
where CallType is not null
group by CallType, Zipcode
order by count desc
```

05-working-with-dataframe - Da × pyspark.sql.GroupedData — PyS ×

community.cloud.databricks.com/?o=946780288864704#notebook/384359114294295/command/3271264990413486

05-working-with-dataframe (Python)

demo-cluster

Cmd 21

```python
fire_df.select("CallType") \
    .where("CallType is not null") \
    .groupBy("CallType") \
    .count() \
    .orderBy("count", ascending=False) \
    .show()
```

(2) Spark Jobs

```
+--------------------+-------+
|            CallType|  count|
+--------------------+-------+
|    Medical Incident|2843475|
|      Structure Fire| 578998|
|              Alarms| 483518|
|   Traffic Collision| 175507|
|Citizen Assist / ...|  65360|
|               Other|  56961|
|        Outside Fire|  51603|
|        Vehicle Fire|  20939|
|        Water Rescue|  20037|
|Gas Leak (Natural...|  17284|
|   Electrical Hazard|  12608|
|Elevator / Escala...|  11851|
|Odor (Strange / U...|  11680|
|Smoke Investigati...|   9796|
|           Fuel Spill|   5198|
```

Transformation:

1. select()
2. where()
3. groupBy()
4. count()
5. orderBy()

Action:

1. show()
2. count()

Functions:

1. display()
2. expr()